nearlearn®

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Course Curriculum

2025

- Latest curriculum
- industry Standard
- Basic to Professional
- All the topics covered

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Data Science & its importance

- KeyElements of DataScience
- Data Warehousing
- Business Intelligence
- Data Visualization
- Data Mining
- Machine Learning
- Artificial Intelligence
- Cloud Computing
- Big Data
- Python Introduction
- Data Structures – int, str, float, bool, complex
- Data Structures – list, tuple, dict and
- Control Flow – if else, for loop, while loop
- Functions
- OOPs
- Packages & Modules
- Types of Variables – Continuous and Categorical
- Interval, Ratio, Ordinal and Nominal
- Terms – Population, Sample, Descriptive Statistics and Inferential Statistics
- Mean, Median and Mode
- Range, Percentile and IQR
- Variance and Standard Deviation
- Correlation and Covariance

- Intro to Numpy Arrays
- Creatingndarrays
- Indexing, Data Processing using Arrays
- Mathematical computing basics
- Basic statistics
- File Input andOutput
- Getting Started with Pandas
- Data Acquisition (Import & Export)
- Selectionand Filtering
- Combining and Merging Data Frames
- Statistical analysis with Pandas Data Frame
- Removing Duplicates & String Manipulation
- Data Cleaning– Handling Missing values
- Data Aggregation withPandas

Visualization in python

- Introduction to Visualization
- Visualization Importance
- Working with Python visualization libraries
- Matplotlib
- Working with Seaborn
  Advance Plots with Seaborn
- Creating Line Plots, Bar Charts, Pie Charts, Histograms,
- Scatter Plots
- Understanding Box plots
- Understanding Probability Distributions, Violin Plots

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- Correlations and Heatmaps
- Regression Problem Analysis
- Mathematical modelling of Regression Model
- Gradient Descent Algorithm
- Use cases
- Model Specification

Data Processing For Machine Learning

- Data Collection & Preparation
- Data Mugging
- Outlier Analysis
- Missing value treatment
- Feature Engineering
- Data Transformation
- Normalization vs Standardization
- Creating Dummies
- Dimensionality Reduction
- Principal Component Analysis

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Linear Regression – Case Study & Project

- Programming Using python
- Building simple Univariate Linear Regression Model
- Multivariate Regression Model
- Modelling of Data
- Variable Significance Identification
- Bifurcate Data into Training / Testing Data set
- Build Model on Training Data Set
- Predict using Testing Data Set
- Validate the Model Performance
- MAPE, RMSE and R2 Value
- Assumptions
- Reason for the Logit Transform
- Logit Transformation
- Hypothesis
- Variable and Model Significance
- Maximum Likelihood Concept
- Log Odds and Interpretation
- Model Specification
- Confusion Matrix
- Accuracy, Recall, Precision and F1 Score
- Forming a Decision Tree Components of Decision Tree Mathematics of Decision Tree Decision Tree Evaluation

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Decision Trees – Random forests & projects

- Practical Examples & Case Study Project
- Credit Card Approval Algorithm Mode

Supervised Machine Learning algorithms

- Linear Regression
- Logistic Regression
- Decision/Classification on Tree Ensemble Models
- Bagging
- Boosting
- Random Forest
- K-Nearest Neighbours (KNN)
- Naive Bayes o Neural Network (Deep Learning)
- Support Vector Machine

Unsupervised Machine Learning algorithms

- Clustering with K-means Clustering
- Bias-Variance Trade off
- Regularization
- Parameter tuning & grid search optimization.

Random Forests

- Random Forest Mathematics
- Examples & use cases using Random Forests

Clustering

- Hierarchical Clustering
- K Means Clustering
- Use Cases for K Means Clustering
- Programming for K Means using Python
- Cluster Size Optimization vs Definition Optimization
- Projects & Case Studies

Principle Component Analysis

- Dimensionality Reduction, Data Compression
- Curse of dimensionality
- Multicollinearity & Factor Analysis
- Concept and Mathematical modelling
- Programming with python
- Neurons, ANN & Working
- Single Layer Perceptron Model
- Multilayer Neural Network
- Feed Forward Neural Network

  Cost Function Formation

  Applying Gradient Descent Algorithm

  Backpropagation Algorithm & Mathematical Modelling

- Programming Flow for backpropagation algorithm
- Use Cases of ANN
- Programming SLNN using Python Programming MLNN using Python

Case Study And Projects

- Real life cases with Python